INCH-POUND
A-A-50437
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COMMERCIAL ITEM DESCRIPTION

EXOTHERMIC CUTTING RODS FOR USE IN PORTABLE EXOTHERMIC CUTTING UNIT (PECU)

The General Services Administration has authorized the use of this commercial item description.

This Commercial Item Description covers the classification of exothermic torch cutting rods which shall be provided for the Portable Exothermic Cutting Unit (PECU), NSN No. 1H 4240-01-272-3841.

Salient characteristics:

DESCRIPTION: Exothermic cutting rods are of tubular construction, made of carbon steel, and are designed to fit into an exothermic torch handle. The exothermic burning reaction begins when pressurized oxygen is passed through the rod and ignited by a spark from a 12 volt ignition source. The ignited oxygen and carbon steel rod consume themselves as the fuel and create a flame (without a welding power supply or explosive fuel gases) that generates temperatures sufficient to cut materials such as steel, stainless steel, aluminum, cast iron and concrete and various non-metallic materials.

1. Sizes. Two sizes of exothermic torch cutting rods shall be provided for the Portable Exothermic Cutting Unit (PECU), NSN No. 1H 4240-01-272-3841.

1/4 (+.003/-.001) inches diameter by 22 (± .25) inches long
6.50 (+.076 -.025) millimeters diameter by 56 (± .5) centimeters long

3/8 (.375/+.003 -.001) inches diameter by 36 inches (± 1/4) long
1 centimenter (+.076 -.025) diameter by 90 (±.5) centimeters long

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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2. Performance Requirements. The oxygen regulator for the Portable Exothermic Cutting Unit (PECU), NSN No. 1H 4240-01-272-3841 is preset at a maximum pressure of 60 pounds per square inch (lb/in^2). The exothermic cutting rods for the PECU shall meet the performance requirements listed below while requiring an oxygen flow rate of not greater than 15 specific cubic feet per minute at a pressure of between 20 lb/in^2 and 60 lb/in^2.

a. Cutting rate. The exothermic cutting rods shall meet the minimum cutting rate in inches per minute (IPM) or quantity of rods, when cutting through the following materials:

MATERIAL	1/4 INCH ROD	3/8 INCH ROD
1/4 INCH MILD STEEL PLATE	32 IPM	18 IPM
1/8 INCH STAINLESS STEEL PLATE	15 IPM	9 IPM
1/2 INCH ALUMINUM PLATE	13 IPM	7 IPM
3 INCH HY-100 STEEL PLATE	Not Reqd	6 inch diameter hole.**

** Quantity of rods not to exceed ten (10) 3/8 inch X 36 inch long.

b. Flame outs. Flame outs shall occur at a rate of not more than one per five rods burned.

c. Burnback. Burnback is defined as when the cutting rod sheath burns faster than the steel fuel on the inside, leaving the steel fuel exposed. Burnback shall occur at a rate of no more than once per five rods burned.

d. Underwater operation. The 3/8 inch diameter cutting rods shall be capable of cutting a 1 inch diameter hole through 1/4 inch thick mild steel plate submerged in water to a depth of 12 inches.

Quality assurance. Sampling for inspection shall be accomplished by the contractor/offeror in accordance with the contract. The government reserves the right to perform any quality inspections or require manufacturer test data, deemed necessary to assure supplies conform to prescribed requirements.

1. Safety requirements: Material Safety Data Sheets (MSDS) in accordance with the Occupational Safety and Health Administration (OSHA) Hazard Communication Standard 29 CFR 1910.1200 shall be provided to point of delivery for each shipment of Exothermic Cutting Rods. Shipments without (MSDS) will be returned.

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Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with public law 94-580 to the maximum extent practical.

Preservation, packaging, packing, and marking.

1. Preservation and packaging: Preservation and packaging shall be as specified in the contract or order.

2. Packing: The exothermic torch cutting rods shall be packed as follows:

a. 1/4 inches diameter X 22 inches long cutting rods. Twenty five (25) 1/4 inches X 22 inches cutting rods shall be packed in a separate container such as a heavy duty cardboard cylinder.

b. 3/8 inches diameter X 36 inches long cutting rods. Twenty five (25) 3/8 inches X 36 inches cutting rods shall be packed in a separate container such as a heavy duty cardboard cylinder.

3. Labeling and marking: In addition to any special marking specified in the contract or order, each container of twenty five (25) 1/4 inches X 22 inches and 3/8 inches X 36 inches cutting rods shall be labeled and marked as follows:

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a. Item description: EXOTHERMIC CUTTING RODS, including the notation, "For use with Portable Exothermic Cutting Unit (PECU) NSN No. 1H 4240-01-272-3841"

b. Size: 1/4 inches diameter X 22 inches long or 3/8 inches diameter X 36 inches long as applicable.

c. National stock number: When no stock number is available, the Commercial and Government Entity Codes (CAGEC) appropriate to the part number shall be shown followed by the manufacturer's name and reference/part number (MFR/PN). The letters "CAGEC' shall be used to identify this information.

Notes.

1. The Government may elect not to evaluate a bid sample from an apparent low bidder for award for this solicitation if a determination of urgency is made by the contracting officer. The bid sample would then be evaluated prior to future solicitation for comparable items.

2. Purchasers should specify the rod size (length and diameter), and specify that the rods be uncoated.

3. The materials shall have no adverse effect on the health of personnel when properly used for their intended purpose.

4. OSHA Standard 29 CFR 1910.1200 should be referred to for Hazard Communication.

CIVIL COORDINATING ACTIVITIES

GSA - FSS

CUSTODIANS

Army - ME
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PREPARING ACTIVITY

Navy - SH
(Project 4240-0563)

REVIEW ACTIVITY

Navy - YD

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